# Axomeme 2.0: Deep Axial Transformer for Ultra-Fast Selection Inference

Architecture & Mathematical Specification

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**Abstract**

**Axomeme 2.0** is a specialized axial attention transformer neural network designed for rapid, site-level evolutionary selection inference on multiple sequence alignments (MSAs) of protein-coding sequences. By replacing traditional Maximum Likelihood (ML) numerical optimization (such as HyPhy MEME or PAML CODEML) with a high-capacity axial transformer backbone and a rank-consistent ordinal CORAL head, Axomeme 2.0 infers Likelihood Ratio Test (LRT) selection statistics and evolutionary parameters in under **10 milliseconds per alignment** (a **4,000× speedup** over classical ML). This specification details the mathematical formulations, network components, attention mechanisms, and decoding algorithms.

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## 1 System Overview

Given an input alignment matrix $\mathbf{X} \in \mathcal{C}^{N \times L}$ of $N$ species (taxa) and $L$ codon sites, Axomeme 2.0 maps sequence variation and phylogenetic distances directly to continuous site-level selection statistics $\hat{y}_{\text{LRT}} \in \mathbb{R}_{\geq 0}$ and selection classifications (Tier 1, Tier 2, or Neutral).

$$f_{\mathbf{\Theta}}(\mathbf{X}, \mathbf{D}) \longrightarrow \left(\hat{y}_{\text{LRT}}, \hat{\alpha}, \hat{\beta}^{+}, \hat{p}^{+}\right) \tag{1}$$

## 2 Input Representation & Dual Tokenization

### 2.1 Dual Codon-AA Tokenization

Each alignment entry $(i, j)$ at species $i$ and codon site $j$ is tokenized across two parallel feature channels:

- **Codon Vocabulary ($|\mathcal{C}| = 64$)**: Sense codons $0 \ldots 63$. Stop codons, gaps, and unknown nucleotides are assigned index 64.
- **Amino Acid Vocabulary ($|\mathcal{A}| = 20$)**: Standard amino acids $0 \ldots 19$. Unknowns / gaps are assigned index 20.

## 2.2 Embedding Layer

$$\mathbf{e}_{\text{codon}} = \mathbf{E}_{\text{codon}}[c_{i,j}] \in \mathbb{R}^{64} \tag{2}$$
$$\mathbf{e}_{\text{aa}} = \mathbf{E}_{\text{aa}}[a_{i,j}] \in \mathbb{R}^{64} \tag{3}$$
$$\mathbf{x}^0_{i,j} = \text{Concat}(\mathbf{e}_{\text{codon}}, \mathbf{e}_{\text{aa}}) + \mathbf{P}_{\text{pos}}[j] + \mathbf{W}_{\text{mds}}\mathbf{m}_i \in \mathbb{R}^{128} \tag{4}$$

where $\mathbf{P}_{\text{pos}} \in \mathbb{R}^{L\times 128}$ is a 1D positional embedding and $\mathbf{m}_i \in \mathbb{R}^4$ represents Multidimensional Scaling (MDS) coordinates derived from the phylogenetic tree distance matrix.

# 3 Axial Transformer Backbone

Axomeme 2.0 incorporates 4 interleaved axial attention blocks to achieve $O(N \cdot L)$ scaling:

## 3.1 Column-Attention Layer (Codon Spatial Context)

Captures local sequence context along the codon dimension $L$:

$$\mathbf{H}_{\text{col}} = \text{MultiHeadAttention}\left(\mathbf{X}, \mathbf{X}, \mathbf{X}\right) \in \mathbb{R}^{(B\cdot N)\times L\times 128} \tag{5}$$

## 3.2 Phylo-Row-Attention Layer (Taxa Context)

Captures phylogenetic variation across species $N$ with distance normalization:

$$\mathbf{D}_{\text{norm}} = \frac{\mathbf{D}}{\bar{D} + 10^{-4}} \tag{6}$$

Pairwise synonymous ($\mathbf{M}_{\text{syn}}$) and non-synonymous ($\mathbf{M}_{\text{nonsyn}}$) transition masks derived from the genetic code are injected directly into the attention logits.

# 4 Multi-Stream Species Representation Pooling

To aggregate representations across $N$ species into a single site vector $\mathbf{h}_j \in \mathbb{R}^{128}$:

$$\mathbf{h}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_{i,j}, \quad \mathbf{h}_{\text{max}} = \max_{i=1}^{N}(\mathbf{x}_{i,j}) \tag{7}$$
$$\mathbf{h}_{\text{attn}} = \sum_{i=1}^{N} \text{Softmax}\left(\mathbf{w}_a^\top \mathbf{x}_{i,j}\right)\mathbf{x}_{i,j}, \quad \mathbf{h}_{\text{diff}} = \text{ReLU}\left(\mathbf{h}_{\text{max}} - \mathbf{h}_{\text{mean}}\right) \tag{8}$$
$$\mathbf{h}_j = \text{Linear}\left(\text{Concat}\left(\mathbf{h}_{\text{mean}}, \mathbf{h}_{\text{max}}, \mathbf{h}_{\text{attn}}, \mathbf{h}_{\text{diff}}\right)\right) \tag{9}$$

# 5 Rank-Consistent CORAL Head & Expected Value Decoder

To guarantee strict rank consistency ($b_0 > b_1 > \cdots > b_{11}$):

$$b_k = b_0 - \sum_{j=1}^{k} \text{Softplus}(\theta_j), \quad z_k = \mathbf{w}^\top \mathbf{h}_j + b_k \tag{10}$$

where $p_k = \sigma(z_k) = P(\text{LRT} > E_{k+1})$.

Continuous predictions are decoded via the **Smooth Expected Value Decoder**:

$$P(\text{Bin } 0) = 1.0 - p_0, \quad P(\text{Bin } m) = p_{m-1} - p_m, \quad P(\text{Bin } 12) = p_{11} \tag{11}$$

$$\hat{y}_{\text{LRT}} = \mathbb{E}[\text{LRT}] = \sum_{m=0}^{12} P(\text{Bin } m)\cdot \mu_m \tag{12}$$

## 6 Performance Summary

| Model / Benchmark | AUC-ROC | AUC-PR | False Positive Rate | Inference Speed |
|---|---|---|---|---|
| HyPhy MEME (Likelihood) | 0.81 | 0.35 | 5.0% | 45 seconds / gene |
| PAML CODEML (Likelihood) | 0.78 | 0.30 | 6.2% | 180 seconds / gene |
| **Axomeme 2.0 (Rebalanced)** | **0.7813** | **0.3253** | **0.46%** | **¡ 10 milliseconds** |

Table 1: Comparison of Axomeme 2.0 against standard Maximum Likelihood selection tools.